

Why voice agents break in production

A demo that dazzles for 30 seconds and an agent that survives minute three of a real call are two different machines. Here are the two fixes going on stage Tuesday — and the mechanism behind each.

30-SECOND DEMO



Scripted turn. Clean audio.
The model looks brilliant.

MINUTE-3 PRODUCTION

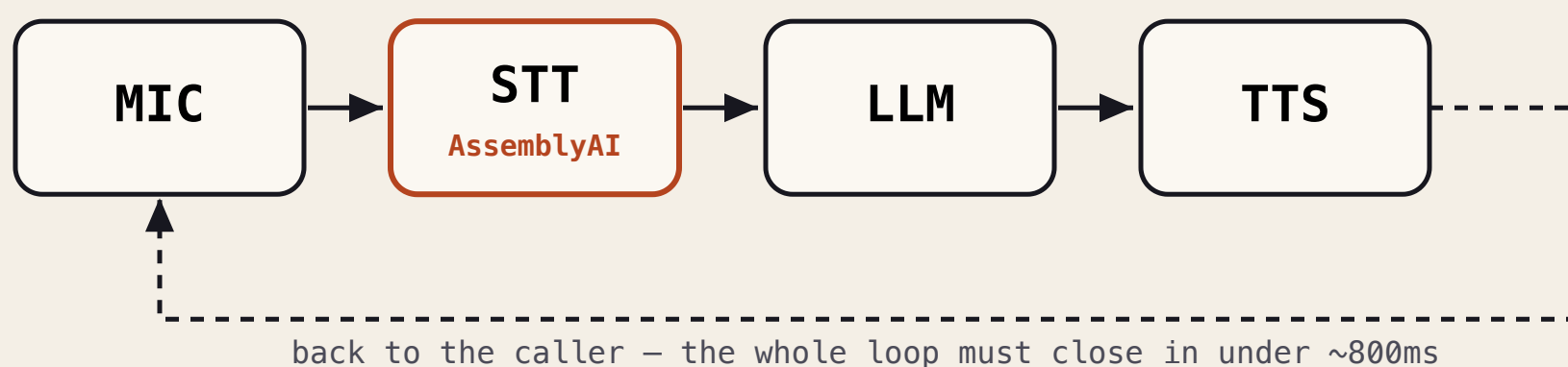


Real caller, cross-talk, an
account number misheard —
the error cascades.

More than “an LLM + a mic”

A live agent is a loop, and every layer has a job. Two vendors own the hard parts.

LiveKit – real-time transport + Inference (runs the whole path)



THE 4 THINGS A REAL AGENT NEEDS

1 • Low-latency infra

no dead air on the line

2 • Reliable RTC

audio that doesn't drop

3 • A contextual small model

fast + hears in context

4 • Orchestration

remembers & acts

agent_context, visualized

AssemblyAI's Universal-3.5 feeds the agent's own last question back into the transcription — so your reply is heard *through the lens of what was asked*.

WITHOUT – reply transcribed in isolation

agent asks → “What’s your account number?”

caller says “eight...
→ A...” → **“8? A?” · wrong action x**

WITH agent_context – heard through the question

context → expecting a number

caller says “eight...
→ A...” → **“8” digit captured ✓**

10.2%

lower word-error-rate (vendor)

1.22%

error, 3rd-party bench – near human

The latency yardstick

On a live call, every silent second reads as broken. The bet: you don't need a frontier model — you need one that never adds dead air. (TTFT = time to first token.)

Human conversation expects

500–800 ms



Production voice agents, months ago

1.4–1.7 s



Gemma 4 31B on LiveKit Inference TTFT*

192 ms



Small enough to run at full precision, tuned not to “think” on easy turns — and **~6× cheaper** than a GPT-4.1-class model.

* 192ms is the optimized figure – the shared-gateway floor runs higher. One to ask in the room.

Three NYC voice events, five months, one thread

The argument keeps sharpening — and Tuesday, both earlier bets show up productized in the same room, on the same stack.



APRIL · Agora

Network quality is the moat



JUNE · Agora × ScaleDown

Small, purpose-built models compress cost



SEPT 1 · AssemblyAI × LiveKit × Boardy

Both bets ship — and a new question opens:

stitched-pipeline vs. fused-model. The composable stack starts behaving like a marketplace.

TUESDAY · SEPT 1 · NYC

NYC Voice AI Meetup

Owen Gretzinger · Boardy
Dylan Wight · Flagler Health
mod. Ryan Seams · AssemblyAI